NAFLD fibrosis prediction: EDA handover and modeling guide

Status as of 17 October 2026. The EDA stage is complete. This document gives the modeling teammate everything needed to continue, and gives the EDA owner the reasoning behind every step.

1. The project in brief

Dataset: Kaggle 'Non-alcohol fatty liver disease (NAFLD)', 605 patients, 62 columns, all with biopsy-confirmed fatty liver.

Goal: predict **significant liver fibrosis (stage 2 or higher)** from routine clinical and lab values, without a biopsy. Task type: binary classification. Benchmark: FIB-4, the standard non-invasive fibrosis score.

The first idea was fatty liver detection (fatty liver vs healthy). It was dropped because the dataset contains no healthy patients (section 2).

2. Why the goal changed

- Steatosis grade counts: grade 1 = 151, grade 2 = 246, grade 3 = 208. There is no grade 0.
- So every patient already has fatty liver. A detector needs both positive and negative examples, and this data has no negatives.
- Fibrosis does vary among these patients (F0 = 196, F1 = 190, F2 = 111, F3 = 84, F4 = 24), so predicting fibrosis severity is possible and clinically useful: it could reduce the need for biopsy.
- Wording to use: everyone in the data has fatty liver, and fibrosis (scarring) varies between patients. Do not say that everyone already has fibrosis.

3. Dataset facts

Item	Value
File	/content/drive/MyDrive/NAFLD_Proj/NAFLD.xlsx, read with pd.read_excel(path, sheet_name='Sheet1')
Sheets	Sheet1 has the data. Sheet2 and Sheet3 are empty

Item	Value
Size	605 rows, 62 columns, 0 duplicate rows
Column 0	Patient No., an ID (never a feature)
Columns 1 to 50	Inputs: demographics, body measures, blood tests, comorbidities
Columns 51 to 61	Biopsy-derived results (never inputs)
Coding	Gender: Female = 1, Male = 2. Most yes/no columns: No = 0, Yes = 1
Original spellings	'Diyabetes Mellitus' and 'Trombosit' (platelets) are the dataset's own spellings. Keep them or code will fail

4. Targets

Target	Short name	Positives	Role
Significant fibrosis (F2 or higher)	SigFibrosis	219 of 605 (36.2%)	Main target
Advanced fibrosis (F3 or higher)	AdvFibrosis	108 (17.9%)	Secondary experiment
Fibrosis group F0-1 / F2 / F3-4	FibGroup	386 / 111 / 108 patients	Recommended second experiment, 3-class, team to confirm
At-risk NASH, recomputed	AtRiskNASH_calc	209 (34.5%)	Short note only

Label checks: SigFibrosis and AdvFibrosis match the raw Fibrosis stage exactly. Cirrhosis has only 24 patients, too few to model.

Class balance: 63.8% vs 36.2%, a ratio of about 1.8 to 1. This is mild imbalance. Use class weights, stratified splits, and PR-AUC and recall next to accuracy. SMOTE is not needed.

The at-risk NASH problem

The dataset's own AtRiskNASH column says it means NAS 4 or higher AND fibrosis 2 or higher. It does not. Recomputing the rule from the raw NAS and Fibrosis columns disagrees with it in 269 of 605 rows (44%). Provided positives = 432, recomputed

positives = 209. In 246 rows it says 1 where the rule says 0 (for example patients with fibrosis 0). It tracks NAS: never 1 for NAS 1 to 3, always 1 for NAS 7 and 8, mixed for NAS 4 to 6. It also does not match the SAF diagnosis. It could not be reconciled, so it is excluded and recomputed as AtRiskNASH_calc.

AtRiskNASH_calc is a strict subset of SigFibrosis and differs from it in only 10 patients (219 vs 209). A full experiment on it would repeat the main one, which is why FibGroup is recommended instead.

5. Leakage control and the final feature set

Never use as inputs: Steatosis, Activity, Fibrosis, NAS, AtRiskNASH, FibrosisStatus, SigFibrosis, AdvFibrosis, Cirrhosis, SAF_Diagnosis, TypeOfDisease, plus the derived AtRiskNASH_calc and FibGroup. Also exclude Patient No. and the EDA-only helper columns n_panel, in_panel and FIB4.

Feature count path:

- 62 columns minus 11 biopsy columns minus the ID = **50 candidate features**
- minus 15 sparse columns (section 6) = **35**
- minus Height and Weight (BMI is kept) = **33 model inputs**: 27 numeric, 5 binary (gender, diabetes, hypertension, hyperlipidemia, metabolic syndrome) and 1 three-level smoking column

Height and Weight are dropped because BMI is computed from them (BMI vs Weight correlation is 0.67, BMI vs Waist 0.76). This is a default. Run an ablation with all five body-size columns and report whether results change.

6. Missing data decisions

Dropped (15 columns, missing 45% or more):

Column	Missing
Microalbumin/Creatinine Ratio	81.0%
CK	74.7%
C Peptide	72.4%
Microalbumin Spot Urine	69.8%
Vitamin D	68.1%
PT	60.8%

Column	Missing
Uric Acid	58.0%
Insulin, HOMA, HOMA insulin-resistance flag	54.2% each
Ceruloplasmin	52.4%
TSH	51.6%
INR	49.8%
Creatinine and BUN	49.6% each

Kept with missing values, imputed inside the pipeline: Total Protein 13.6%, LDH 7.3%, HbA1c 6.8%, Waist 4.8%, Smoking 4.6%, Ferritin 2.8%, Mean Platelet Volume 2.3%, Hip 1.2%, and a few under 1%. Numeric: median. Binary and smoking: most frequent.

Why the threshold is 45% and not 50%

Missingness is not random. Patients who got an extended lab panel had much higher fibrosis rates than those who did not:

Test	Fibrosis rate if tested	If not tested
Vitamin D	59.6%	25.2%
PT	56.5%	23.1%
TSH	48.1%	25.0%
INR	47.0%	25.2%
BUN and Creatinine	46.9%	25.3%
Insulin	44.4%	29.3%
LDH (control, not informative)	36.4%	34.1%

INR, BUN and Creatinine sit just under 50% missing but behave exactly like the dropped columns, so the cutoff was set at 45%. The threshold was chosen after looking at the data, and the report should say so.

The panel

The flags for INR, BUN, Creatinine, TSH, Ceruloplasmin, Insulin and Uric Acid are almost perfectly correlated (0.83 to 1.00 with the INR/BUN/Creatinine group). Count of panel tests per patient (n_panel): 0 tests = 299 patients (fibrosis rate 25.1%), 3 to 5 tests = 23

patients, 6 tests = 58 (44.8%), 7 tests = 225 (46.2%). in_panel = 1 when n_panel is 5 or more: 297 patients, versus 308 with in_panel = 0. Fibrosis 2 or higher is about 25% without the panel and about 48% with it.

The kept features barely differ between the two groups (median age 47 vs 47, AST 43 vs 41, ALT 67 vs 65, platelets 244 vs 232). So the 33-feature model is largely free of referral information, but performance must still be reported separately for each group.

Consequences: no missing-value indicator flags in the main model, and stratify splits on target plus in_panel.

7. What the EDA found

Nine numeric features separate the classes after multiple-testing correction. Medians shown as no fibrosis to fibrosis:

Feature	Median change	Single-feature AUC	p-value
Age	44 to 51	0.647	2e-9
HbA1c	5.70 to 6.00	0.641	2e-8
Glucose	98 to 105	0.624	5e-7
Platelets (Trombosit)	247.5 to 220.5 (lower)	0.618 as a lower-is-riskier marker	2e-6
GGT	46 to 56	0.604	2e-5
BMI	30.8 to 32.4	0.600	5e-5
Hip circumference	103 to 106	0.592	2e-4
Mean platelet volume	8.6 to 8.9	0.591	3e-4
AST	39 to 44	0.589	3e-4

Direction warning: in the notebook output the AUC column is computed so that a value below 0.5 means HIGHER in the fibrosis group. The table above shows the usual orientation. Always read direction from the medians.

Coded features (unadjusted):

Feature	Fibrosis rate, No to Yes	Odds ratio
Diabetes (225 patients have it)	25% to 55%	about 3.6
Hypertension	31% to 46%	about 1.9

Feature	Fibrosis rate, No to Yes	Odds ratio
Metabolic syndrome	29% to 40%	about 1.6
Hyperlipidemia	31% to 40%	about 1.5
Gender	Female 40%, Male 32%	about 1.4 (female vs male)

Diabetes is the strongest single feature. These are associations, not causes, and diabetes overlaps with glucose, HbA1c and age.

Redundancy (Spearman): AST and ALT 0.84, Total Cholesterol and LDL 0.81, BMI and Waist 0.76, BMI and Hip 0.75, Weight and Hip 0.74. All below 0.85, so mild. Tree models ignore it; logistic regression handles it with regularization.

Data quality: BMI matches weight over height squared in 602 of 605 rows. Three rows differ by more than 1 point (patients 61, 429, 520; largest gap 2.89). Recomputing BMI from height and weight is recommended; see open decisions.

Take-away: every single feature is modest (best about 0.65), so the signal is spread across many weak features. Expect a modest combined result.

8. FIB-4 benchmark

Formula: FIB-4 = (Age x AST) / (Platelets x square root of ALT). Platelet units are right (FIB-4 median 1.04, range 0.26 to 6.30). 604 patients have a value (one missing platelet count).

Measure	Significant fibrosis	Advanced fibrosis
AUC (95% CI)	0.703 (0.660 to 0.744)	0.767 (0.718 to 0.814)
Cutoff 1.3: sensitivity / specificity	0.50 / 0.78	0.66 / 0.75
Cutoff 1.3: share flagged	32%	32%
Cutoff 2.67: sensitivity / specificity	0.11 / 0.99	0.18 / 0.98
Cutoff 2.67: share flagged	4%	4%

FIB-4 misses half of significant fibrosis at its usual cutoff, so there is room to improve sensitivity. FIB-4 is derived from Age, AST, ALT and platelets, so it must never be a model input.

9. Notebook variables to know

Name	What it is
df	Raw Sheet1, untouched
data	Working copy with short names, derived columns and helper columns
biopsy_cols	The 13 columns that must never be inputs (11 original plus AtRiskNASH_calc and FibGroup)
feature_cols	The 50 candidate inputs
miss	Missing fraction per candidate feature
drop_sparse	The 15 dropped sparse columns
model_features	The 35 kept columns
cat_like	Columns with 3 or fewer unique values (the coded ones)
n_panel, in_panel	Extended panel count and flag. EDA and stratification only
FIB4	Benchmark score. Never an input
feats	Final 33 model inputs
num_feats, bin_cols, smoke	Column groups used by the preprocessing pipeline
strata	Target plus in_panel as text, used to stratify splits
SEED	42, used everywhere

10. Setup cell (run this first, every session)

Colab forgets variables when the runtime restarts or cells run out of order, which causes NameError. One cell rebuilds everything.

```
import pandas as pd, numpy as np
from google.colab import drive
drive.mount('/content/drive')

path = '/content/drive/MyDrive/NAFLD_Proj/NAFLD.xlsx'
df = pd.read_excel(path, sheet_name='Sheet1')
data = df.copy().rename(columns={
```

```
 'Significant Fibrosis (No=0, Yes=1) (If Fibrosis 2 and above, there is
Significant Fibrosis)': 'SigFibrosis',
 'Advanced Fibrosis (No=0, Yes=1) (If Fibrosis is 3 and above, there is
Advanced Fibrosis)': 'AdvFibrosis',
 'NAS score>=4 and Fibrosis>=2 (If NAS score>=4 and Fibrosis>=2 = 1, If not =
0)': 'AtRiskNASH',
 'Cirrhosis (No=0, Yes=1) (If Fibrosis is 4, there is Cirrhosis)':
'Cirrhosis',
 'Fibrosis status (No=0, Yes=1) (Fibrosis 1 and above, there is Fibrosis)':
'FibrosisStatus',
 'NAS score according to Kleiner': 'NAS',
 'Diagnosis according to SAF (NASH=1, NAFL=2)': 'SAF_Diagnosis',
 'Type of Disease (Mild illness=1, Severe illness=2)': 'TypeOfDisease'})

data['AtRiskNASH_calc'] = ((data['NAS'] >= 4) & (data['Fibrosis'] >=
2)).astype(int)
data['FibGroup'] = np.select([data['Fibrosis'] <= 1, data['Fibrosis'] == 2],
[0, 1], default=2)

biopsy_cols = ['Steatosis', 'Activity', 'Fibrosis', 'NAS', 'AtRiskNASH',
'AtRiskNASH_calc', 'FibrosisStatus',
               'SigFibrosis', 'AdvFibrosis', 'Cirrhosis', 'SAF_Diagnosis',
'TypeOfDisease', 'FibGroup']
feature_cols = [c for c in data.columns if c not in biopsy_cols + ['Patient
No.']]
print(len(feature_cols))                    # expect 50

miss = data[feature_cols].isna().mean()
drop_sparse = miss[miss > 0.45].index.tolist()
model_features = [c for c in feature_cols if c not in drop_sparse]
print(len(drop_sparse), len(model_features))  # expect 15, 35

# helper columns (created AFTER feature_cols so they never become features)
panel = ['INR', 'BUN', 'Creatinine', 'TSH', 'Ceruloplasmin', 'Insulin', 'Uric
Acid']
data['n_panel'] = data[panel].notna().sum(axis=1)
data['in_panel'] = (data['n_panel'] >= 5).astype(int)
data['FIB4'] = (data['Age'] * data['AST']) / (data['Trombosit'] *
np.sqrt(data['ALT']))

# optional: fix the 3 inconsistent BMI rows
data['Body Mass Index'] = (data['Weight'] / (data['Height'] / 100) **
2).round(1)

SEED = 42
smoke = 'Smoking Status (Never Smoked=1, Left Smoking=2, Smoking=3)'
```

```
cat_like = [c for c in model_features if data[c].nunique() <= 3]
feats = [c for c in model_features if c not in ['Height', 'Weight']]
bin_cols = [c for c in cat_like if c in feats and c != smoke]
num_feats = [c for c in feats if c not in bin_cols + [smoke]]
print(len(feats), len(num_feats), len(bin_cols))   # expect 33, 27, 5

X, y = data[feats], data['SigFibrosis']
strata = y.astype(str) + '_' + data['in_panel'].astype(str)
```

11. Modeling tasks

M2, hold out a test set once, stratified on target and panel group. Nobody looks at it again until the final evaluation. Expect 484 training and 121 test patients, about 36% positives in both.

M3 to M5, build the pipeline, run repeated CV, then compare with FIB-4 on the same patients:

```
from sklearn.model_selection import train_test_split,
RepeatedStratifiedKFold, StratifiedKFold, cross_validate, cross_val_predict
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import roc_auc_score
from xgboost import XGBClassifier

X_train, X_test, y_train, y_test, s_train, s_test = train_test_split(
    X, y, strata, test_size=0.2, stratify=strata, random_state=SEED)

pre = ColumnTransformer([
    ('num', Pipeline([('imp', SimpleImputer(strategy='median')), ('sc',
StandardScaler())]), num_feats),
    ('bin', SimpleImputer(strategy='most_frequent'), bin_cols),
    ('smk', Pipeline([('imp', SimpleImputer(strategy='most_frequent')),
                      ('oh', OneHotEncoder(handle_unknown='ignore'))]),
[smoke]),
])   # no missing-value indicators on purpose

spw = (y_train == 0).sum() / (y_train == 1).sum()
models = {
  'LogReg': LogisticRegression(max_iter=3000, class_weight='balanced'),
  'RandomForest': RandomForestClassifier(n_estimators=500,
```

```
min_samples_leaf=5, class_weight='balanced', random_state=SEED, n_jobs=-1),
  'XGBoost': XGBClassifier(n_estimators=300, max_depth=3, learning_rate=0.05,
subsample=0.8, colsample_bytree=0.8,
                           scale_pos_weight=spw, eval_metric='logloss',
random_state=SEED),
}

cv = list(RepeatedStratifiedKFold(n_splits=5, n_repeats=5,
random_state=SEED).split(X_train, s_train))
scoring = ['roc_auc', 'average_precision', 'recall', 'precision', 'f1']
summary = {}
for name, m in models.items():
    r = cross_validate(Pipeline([('pre', pre), ('model', m)]), X_train,
y_train, cv=cv, scoring=scoring, n_jobs=-1)
    summary[name] = {s: f"{r['test_' + s].mean():.3f} +/- {r['test_' +
s].std():.3f}" for s in scoring}
print(pd.DataFrame(summary).T)

# M5: same patients as FIB-4
skf = list(StratifiedKFold(n_splits=5, shuffle=True,
random_state=SEED).split(X_train, s_train))
fib4 = data.loc[X_train.index, 'FIB4']; ok = fib4.notna().values; yt =
y_train.values
print('FIB-4 AUC on train rows:', round(roc_auc_score(yt[ok],
fib4.values[ok]), 3))
for n, m in models.items():
    p = cross_val_predict(Pipeline([('pre', pre), ('model', m)]), X_train,
y_train, cv=skf, method='predict_proba')[:, 1]
    print(n, 'AUC:', round(roc_auc_score(yt[ok], p[ok]), 3))
```

Then: paired bootstrap of the AUC difference against FIB-4, tune the best model, choose a decision threshold for sensitivity, SHAP for explainability, performance split by in_panel, and one final evaluation on the test set.

12. Rules, expectations and red flags

- Do not use the test set for any decision. Evaluate it once at the end.
- Do not use any biopsy column, FIB4, n_panel or in_panel as an input.
- Keep preprocessing inside the pipeline so imputation and scaling are fitted only on training folds.
- Report mean and spread across folds, and confidence intervals for the final comparison.

- Realistic target: AUC around 0.70 to 0.78 against FIB-4 at 0.703. A gain in sensitivity at similar specificity is the strongest clinical story.
- Red flag: AUC above about 0.9. Stop and look for leakage.
- Also report results separately for in_panel = 0 and in_panel = 1.

13. Decisions that changed during the EDA

- INR, BUN and Creatinine were first planned to be kept (under 50% missing). They were dropped after the missingness analysis showed they were part of the panel.
- AtRiskNASH was planned as a secondary experiment. It was replaced by FibGroup as the recommendation after the label check failed.
- The AUC direction in the feature ranking table was first described the wrong way round. Below 0.5 means higher in the fibrosis group.
- BMI and Weight are correlated at 0.67, not very high. Dropping Height and Weight is a default choice, not a necessity.
- The kept feature set has 6 coded columns (gender, diabetes, hypertension, hyperlipidemia, metabolic syndrome, smoking), not 4.

14. Open decisions for the team

- Second secondary experiment: FibGroup (recommended) or AtRiskNASH_calc as a note only.
- Recompute BMI for the three inconsistent rows, or leave the source untouched and report them.
- Body-size feature ablation (BMI, Waist, Hip vs all five columns).
- Optional exploratory subset model on the roughly 283 patients with a near-complete panel (insulin and HOMA available). Different, sicker population, so never compare it directly with the main model.
- Whether to include the 15 dropped columns in a sensitivity run.

15. Questions you may be asked

Why not fatty liver detection? The dataset has no healthy patients; steatosis grades are only 1, 2 and 3. A classifier needs negatives.

Why significant fibrosis? Clinically important threshold, 36% positive so classes are workable, 219 positives is enough for cross-validation, and FIB-4 gives a benchmark. Advanced fibrosis is 18% and cirrhosis only 24 patients.

What is leakage here? Any column that contains or is derived from the answer. All biopsy columns, the fibrosis flags, FIB-4 (built from the same labs) and the panel counters are excluded.

Why not SMOTE? Imbalance is mild (1.8 to 1). Class weights and stratification are enough, and synthetic samples add risk with only 605 patients.

Why drop 15 columns and why 45%? Those tests were ordered mostly for sicker patients, so having the test is itself a signal. The cutoff was set after INR, BUN and Creatinine behaved like the panel.

Why compare with FIB-4? It is the standard non-invasive score. It reaches AUC 0.703 here, with sensitivity only 0.50 at its usual cutoff.

Why AUC and recall and not accuracy? Always predicting no fibrosis gives 63.8% accuracy. AUC, PR-AUC and recall show whether the model finds patients with fibrosis.

Why is the expected performance modest? The best single feature reaches about 0.65 and FIB-4 about 0.70. Routine labs carry limited fibrosis signal.

Why recompute AtRiskNASH? The provided column disagreed with its own definition in 44% of rows and followed NAS rather than fibrosis.

Why stratify on in_panel too? The panel group has about twice the fibrosis rate, so each fold should contain both groups in the right proportions.

16. Glossary

- **AUC (ROC-AUC):** probability that a random patient with fibrosis gets a higher score than one without. 0.5 is chance, 1.0 is perfect.
- **PR-AUC:** area under the precision-recall curve. Chance level equals the positive rate (about 0.36).
- **Sensitivity (recall):** share of true fibrosis cases the model finds.
- **Specificity:** share of non-fibrosis patients correctly cleared.
- **Precision:** share of flagged patients who truly have fibrosis.
- **Stratified split:** keeps the same class proportions in every part.
- **Cross-validation:** repeated train and validate on different folds to get a stable score.
- **Leakage:** information in the inputs that reveals the answer, making scores unrealistically high.
- **Imputation:** filling missing values (median for numbers, most frequent for categories).
- **SHAP:** method that shows how much each feature pushed a prediction up or down.
- **Mann-Whitney U:** rank test for whether a feature differs between two groups, without assuming normality.

- **Odds ratio:** how many times higher the odds of fibrosis are in one group than another.